

DARSH THAKUR

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Education

Shiv Nadar Institute of Eminence, Greater Noida

Aug 2024 – May 2028

B.Tech, Electrical and Computer Science Engineering

Experience

AI/ML Engineering Intern — Siemens

Summer 2025

- Built an internal AI task-management tool (RAG + MCP) that auto-decomposes projects into granular subtasks with estimated human-hours.
- Engineered team-aware assignment logic factoring skill profiles, strengths, and workload capacity for balanced distribution.
- Scoped Slack/Jira integration for real-time syncing; designed the API and auth workflows before project handoff.

Tech Intern — WarpVelocity

2025

- Built an enterprise insurance-document analyzer (RAG + Flask API) for keyword querying of arbitrarily large policy documents at configurable detail levels.
- Engineered end-to-end ingestion: OCR extraction, tokenization, sandboxed validation for uncapped uploads, and context-aware retrieval via Streamlit.

Projects

Hound Suite — Three From-Scratch Security CLIs

Rust | 2026

- **leakhound**: scans git history for committed secrets by parsing the `.git` object database directly (packfiles, delta chains, refs, commit graph) in ~1,500 lines with no `git` binary or `libgit2`; recovers dangling blobs invisible to `git log`.
- **credhound**: enumerates on-disk credentials (SSH keys, cloud creds, tokens) and reasons about *who else can read them* via hand-written `/proc` reads and Unix permission modeling.
- **roundhound**: rotation enforcer that ages credentials against policy and auto-rotates SSH keys to fresh Ed25519 pairs on a systemd timer, capping exposure windows.

The Paladin — Local-First Password Manager

Python | 2026

- Encrypts a single local vault file behind one master password — no cloud, no daemon, no browser. Published to PyPI (`pipx install basic-password-manager`).

spider-base — Headless Browser Automation Framework

Playwright | 2026

- Built a Playwright automation base running 5 unattended bots (ERP scraping, attendance, course registration) on a self-hosted Linux homelab.
- Engineered resilience for unsupervised runs: screenshot-on-failure with mobile push alerts, stale-selector detection, and headless rendering-parity fixes.

GA-Based CPU Scheduling in xv6 — Research Paper

C, RISC-V | 2026

- Co-authored a paper on a genetic-algorithm non-preemptive scheduler in xv6; extended the kernel with custom system calls. GA beat SJF by 2.9% and SRTF by 6.3% on combined cost.

Multi-Mode USB Rubber Ducky — Embedded HID Attack Device

Embedded C, STM32 | 2025

- Developed a multi-payload HID device on STM32 with hardware buttons for payload switching (4–6 payloads) and force start/stop of keystroke injection.

Leadership & Activities

Core Coordinator — HackData 2026, National Hackathon (SNU)

2025–2026

- Ran a 24-hour national hackathon: 2,000+ registrations, 250 on-campus finalists across Hackathon, CP Duels, and CTF tracks; INR 3 lakh+ prize pool sponsored by HP. Student judge for the top 10 teams.

Co-Founder & Coordinator — Aeronautics Club (SNU)

2025–2026

- Founded the campus aeronautics club; led 3 RC-plane builds and ran KSP rocket-design events with audience-voted configurations.

Head of POCs — Cultural Committee (SNU)

2026–Present

- Primary liaison for 50+ clubs and societies: inter-club coordination, query resolution, and logistics for large-scale campus events.

Technical Skills

Languages: Rust, Python, C, Embedded C, JavaScript, MATLAB

Systems & Security: Linux internals, `/proc`, git object-database parsing, Unix file-permission modeling, SSH key rotation (Ed25519), authenticated encryption, TLS/JA3, HID attacks, CompTIA Security+ (in progress)

AI/ML: RAG, MCP, LLM APIs, Scikit-learn, Optuna, Gradient Boosting, Genetic Algorithms

Tools & Infra: Git, Flask, Streamlit, Playwright, Tesseract OCR, systemd, STM32, xv6, self-hosted Linux homelab, PyPI/pipx, Jira, Slack API